

NTMS Terraform D-2

25 Questions

1. In Terraform, how can a dependency on a provider be established? (select three)

- 0/0 ☐ A Adding the provider's documentation URL as a comment in the Terraform code.
- 0/0 ☐ B Existence of any provider plugins found locally in the working directory.
- 0/0 ☒ C Using a resource or data block that belongs to that provider in the configuration.
- 0/0 ☒ D Declaring a provider block in the configuration, including any required version constraints.
- 0/0 ☒ E Having existing resource instances for that provider recorded in the current state.

2. You added resources from a new provider to an existing configuration and the next terraform plan results in an error regarding the provider. What should you do first?

- 0/0 ☐ A since a new provider has been introduced, terraform validate should be run to download the new plugin and verify the config
- 0/0 ☐ B since a new provider has been introduced, adding it to required_providers is sufficient and running terraform init isn't needed
- 0/0 ☐ C you can't mix resources from different providers within the same configuration file, so Henry should create a module for the new resource and reference it from the main configuration
- 0/0 ☒ D since a new provider has been introduced, terraform init needs to be run to download the new plugin

3. You configure two aws providers in the same module, as shown below, and Terraform returns Error: Duplicate provider configuration and says to set an additional argument for alternative configurations.

Which argument must you add to the second provider block so both can be used?

```
provider "aws" {  
  region = "us-east-1"  
}  
  
provider "aws" {  
  region = "ap-south-1"  
}
```

- 0/0 ☐ A version
- 0/0 ☐ B profile
- 0/0 ☒ C alias
- 0/0 ☐ D locals

4. Your team is discussing why Terraform maintains a state file. Which of the following are valid reasons why Terraform requires state? (select three)

- 0/0 ☐ A encrypt sensitive data in your configuration files
- 0/0 ☒ B improve performance by caching resource attributes
- 0/0 ☒ C map real-world resources to your Terraform configuration
- 0/0 ☐ D validate provider credentials before applying changes
- 0/0 ☐ E automatically fix HCL configuration syntax errors
- 0/0 ☒ F track metadata such as resource dependencies

5. You are reviewing Terraform code from a colleague and discover a .terraform directory. What is the purpose of this directory?

- 0/0 ☐ A The .terraform directory contains the validated Terraform configuration and plan files used during terraform apply.
- 0/0 ☐ B The .terraform directory is where Terraform saves local state files and state backups by default.
- 0/0 ☒ C The .terraform directory stores Terraform's local working data, including installed provider and module plugins, backend metadata, and state lock information.
- 0/0 ☐ D The .terraform directory holds cached provider credentials and sensitive variables used for authentication.

6. You run terraform init in a new working directory. The output shows Terraform downloading the aws and time provider plugins. On this machine, where does Terraform store those provider plugins?

- 0/0 ☐ A The .terraform/plugins directory in the current working directory
- 0/0 ☐ B The .terraform.d directory in the current working directory
- 0/0 ☒ C The .terraform/providers directory in the current working directory
- 0/0 ☐ D /etc/terraform/plugins

7. How do you specify which provider Terraform should install for a configuration?

- 0/0 ☒ A Define the provider in required_providers and add a matching provider block in the configuration.
- 0/0 ☐ B Run terraform apply first so Terraform can detect and install the correct provider automatically.
- 0/0 ☐ C Set the provider name in a variable and pass it with -var when running terraform init.
- 0/0 ☐ D Download the provider binary and place it in the working directory with your Terraform configuration files.

8. Why should a user specify provider version constraints in their Terraform configuration?

- 0/0 ☐ A to remove older versions of the provider
- 0/0 ☒ B providers are released on a separate schedule from Terraform itself; therefore, a newer version could introduce breaking changes
- 0/0 ☐ C to ensure that the provider version matches the version of Terraform you are using
- 0/0 ☐ D to match the version number of your application being deployed via Terraform

9. Which of the following is considered a Terraform plugin?

- 0/0 ☐ A module
- 0/0 ☐ B backend
- 0/0 ☒ C provider
- 0/0 ☐ D variable

10. You run terraform plan in a workspace that has existing infrastructure. Terraform shows that it will update 3 resources, create 2 new resources, and destroy 1 resource. How does Terraform determine what changes need to be made?

- 0/0 ☐ A Terraform uses specific cloud provider services to identify resources that have drifted
- 0/0 ☒ B Terraform compares the desired state in the configuration with the current state in the state file to build the plan
- 0/0 ☐ C Terraform reads the previous plan output file and applies any changes that weren't completed
- 0/0 ☐ D Terraform queries the cloud provider APIs to discover all resources and compares them to the configuration file

11. True or False? Multiple providers can be declared within a single Terraform configuration file.

- 0/0 ☒ T True
- 0/0 ☐ F False

12. Before a new Terraform provider can be used in a configuration, what steps are required? (select two)

- 0/0 ☒ A Declare and configure the provider in the configuration with the required arguments.
- 0/0 ☒ B Initialize the working directory using terraform init to download and install the provider.
- 0/0 ☐ C Add the provider name and configuration to "[terraform.tfvars](#)".
[terraform.tfvars](#) and pass it with -var .
- 0/0 ☐ D Run terraform validate to compile the configuration and fetch the plugin from the Terraform registry.

13. Which of the following best describes a Terraform provider?

- 0/0 ☒ **A** a plugin that Terraform uses to translate the API interactions with the service or provider
- 0/0 ☐ **B** describes an infrastructure object, such as a virtual network, compute instance, or other components
- 0/0 ☐ **C** serves as a parameter for a Terraform module that allows a module to be customized
- 0/0 ☐ **D** a container for multiple resources that are used together

14. In the top-level terraform block, which setting specifies a provider's source and version constraints?

- 0/0 ☒ **A** required_providers
- 0/0 ☐ **B** backend
- 0/0 ☐ **C** required_version
- 0/0 ☐ **D** provider

15. A provider alias is used for what purpose in a Terraform configuration file?

- 0/0 ☐ **A** sharing a provider configuration from a child module back to the root
- 0/0 ☒ **B** defining multiple configurations of the same provider and binding resources to them
- 0/0 ☐ **C** to signify what resources should be deployed to a certain cloud provider
- 0/0 ☐ **D** selecting the default workspace for a provider during planning

16. Your team wants Terraform to manage a new SaaS with just a REST API. Which Terraform component defines resource schemas, manages authentication and interactions with the API, and provides resources and data sources for configurations?

- 0/0 ☐ **A** Terraform backend
- 0/0 ☒ **B** Terraform provider
- 0/0 ☐ **C** Terraform module

17. You have a Terraform configuration file defining resources to deploy on VMware, yet there is no related state file. You have successfully run terraform init already. What happens when you run a terraform apply?

- 0/0 ☐ **A** Terraform will produce an error since there is no state file
- 0/0 ☐ **B** All existing infrastructure on VMware will be deleted, and the resources defined in the configuration file will be created.
- 0/0 ☐ **C** Since there is no state file associated with this configuration file, the defined resources will not be created on the VMware infrastructure.
- 0/0 ☒ **D** Terraform will create the resources defined in the configuration file and write a new state file.

18. Given the example code below, where will S3 buckets named logs and operations be created?

```
provider "aws" {  
  region = "us-east-1"  
}  
  
provider "aws" {  
  alias = "west"  
  region = "us-west-2"  
}  
  
provider "aws" {  
  alias = "eu"  
  region = "eu-west-2"  
}  
  
resource "aws_s3_bucket" "logs" {  
  <bucket arguments>  
}  
  
resource "aws_s3_bucket" "operations" {  
  provider = aws.west  
  <bucket arguments>  
}  
  
resource "aws_s3_bucket" "config" {  
  provider = aws.eu  
  <bucket arguments>  
}
```

- 0/0 ☐ A Neither bucket will be created since provider is not a valid argument for a resource block
- 0/0 ☒ B logs: us-east-1
operations: us-west-2
- 0/0 ☐ C logswill not be created since no provider was indicated
operations: us-west-2

19. A project is using version 6.0.2 of the hashicorp/aws provider. The configuration's version constraint is set to ~> 6.0.0. Two newer versions are available, 6.0.5 and 6.1.0. What is the effect of running terraform init -upgrade<?

- 0/0 ☐ A The command will fail because a provider is already installed.
- 0/0 ☐ B It will keep version 6.0.2 because the constraint isn't applicable to the latest version of the provider.
- 0/0 ☐ C It will use the ~> 6.0.0 version constraint and install the latest hashicorp/aws provider, version 6.1.0.
- 0/0 ☒ D It will install the newest provider version that matches the ~> 6.0.0 constraint, which is 6.0.5.

20. What are the primary benefits of Terraform state? (select three)

- 0/0** **A** allows Terraform to automatically update your infrastructure in real-time by continuously monitoring your cloud provider for changes without needing to run a plan or apply
- 0/0** **B** tracks the current state of your infrastructure, enabling it to understand what resources exist and need to change
- 0/0** **C** maintains relationships between resources, helping Terraform understand dependencies and ensure proper resource ordering
- 0/0** **D** allows multiple team members to collaborate safely by tracking who is making changes and preventing concurrent modifications

21. Which statements describe the key benefits that Terraform providers offer to users? (select three)

- 0/0** **A** Manage storage, encryption, and locking of Terraform state files across remote backends and workspaces.
- 0/0** **B** Extend Terraform via a plugin model so new services/features can ship without changing Terraform Core
- 0/0** **C** Abstract the target platform's API behind a consistent resource/data-source schema and lifecycle
- 0/0** **D** Enable one workflow to manage resources across many platforms by plugging in different providers

22. You must ensure Terraform always installs the same Azure provider release during initialization. Which configuration correctly pins the provider?

- 0/0** **A**

```
data "azurerm" {  
  source = "hashicorp/azurerm"  
  version = "4.51.0"  
}
```
- 0/0** **B**

```
required_providers {  
  azurerm = {  
    source = "hashicorp/azurerm"  
    version = "4.51.0"  
  }  
}
```
- 0/0** **C**

```
terraform {  
  required_providers {  
    azurerm = {  
      source = "hashicorp/azurerm"  
      version = "4.51.0"  
    }  
  }  
}
```

23. You need to deploy resources across two different Azure subscriptions in the same Terraform configuration. How do you configure Terraform to handle this?

- 0/0 **A** Use the default provider for one account and modules for the second Azure account to deploy resources.
- 0/0 **B** Create two separate Terraform configurations in different directories, one for each account.
- 0/0 **C** Define multiple provider blocks with different alias attributes and reference them in resources.
- 0/0 **D** Install two different versions of the Azure provider using version constraints and point each one to a different provider.

24. One of your team members manually deleted an EC2 instance through the AWS console that Terraform was managing. What happens when you run terraform plan?

- 0/0 **A** Terraform removes the missing instance from the state file automatically
- 0/0 **B** Terraform detects the missing instance and shows that it will recreate the resource
- 0/0 **C** Terraform fails with an error saying the missing resource cannot be found
- 0/0 **D** Terraform ignores the missing instance and shows no changes needed

25. True or False? In most cases, you can migrate Terraform state between supported backends at any time, even after running your first terraform apply.

- 0/0 **T** True
- 0/0 **F** False